n) Unusually strong magnetic fields. It should be noted that solar magnetic disturbances might result in the flow of telluric currents in voltage regulator neutrals.

o) Parallel operation. It should be noted that while parallel operation is not unusual, it is desirable that users advise the manufacturer if paralleling with other voltage regulators is planned, and the characteristics of the transformers or reactors so involved.

4.3.3.1 Control

The control, depending on its construction, may be sensitive to altitude considerations. The manufacturer should be consulted where applications exceed 2000 m (6600 ft). The control shall withstand an altitude of up to 3000 m (10 000 ft) without loss of control.

4.4 Frequency

Step-voltage regulators shall be designed for operation at a frequency of 60 Hz or 50 Hz as specified.

5. Rating data

5.1 Cooling classes of voltage regulators

Voltage regulators shall be identified according to the cooling method employed. For liquid-immersed voltage regulators, this identification is expressed by a four-letter code as described in 5.1.1 through 5.1.4.

5.1.1 Liquid-immersed (fire point ≤ 300 °C) air-cooled

Internal cooling medium in contact with the windings is insulating liquid with fire point ≤ 300 °C:

a) Liquid-immersed self-cooled (Class ONAN).

b) Liquid-immersed self-cooled/forced-air-cooled (Class ONAN/ONAF).

5.1.2 Liquid-immersed (fire point > 300 °C) air-cooled

Internal cooling medium in contact with the windings is insulating liquid with fire point > 300 °C:

a) Liquid-immersed self-cooled (Class KNAN).

b) Liquid-immersed self-cooled/forced-air-cooled (Class KNAN/KNAF).

5.1.3 Liquid-immersed (fire point ≤ 300 °C) water-cooled

Internal cooling medium in contact with the windings is insulating liquid with fire point ≤ 300 °C:

a) Liquid-immersed water-cooled (Class ONWF).

b) Liquid-immersed water-cooled/self-cooled (Class ONWF/ONAN).

5.1.4 Liquid-immersed (fire point > 300 °C) water-cooled

Internal cooling medium in contact with the windings is insulating liquid with fire point > 300 °C:

a) Liquid-immersed water-cooled (Class KNWF).

b) Liquid-immersed water-cooled/self-cooled (Class KNWF/KNAN).

5.2 Ratings

Ratings for step-voltage regulators are continuous and based on not exceeding the temperature limits covered in Table 3. Other winding rises may be recognized for unusual ambient conditions specified.

Ratings covered by this standard shall be expressed in the terms given in 5.2.1 and as specified in 5.2.2.

Table 2—Limits of temperature rise

Item	Type of apparatus [a]	Winding temperature rise by resistance, °C	Hottest-spot winding temperature rise, °C
(1)	55 °C rise liquid immersed	55	65
	65 °C rise liquid immersed	65	80
(2)	Metallic parts in contact with current-carrying conductor insulation shall not attain a temperature rise in excess of the winding hottest-spot temperature rise.		
(3)	Metallic parts other than those described in item (2) shall not attain excessive temperature rises at maximum rated load.		
(4)	The temperature rise of the insulating fluid shall not exceed 55 °C (55 °C rise unit) or 65 °C (65 °C rise unit) when measured near the surface of the fluid.		

[a] Apparatus with specified temperature rise shall have an insulation system that has been proven by experience, general acceptance, or an accepted test.

5.2.1 Terms in which rating is expressed

The rating of a step-voltage regulator shall be expressed in the following terms:

a) Kilovoltampere (kVA)

b) Number of phases

c) Frequency

d) Voltage

e) Current

f) Voltage range in percent (raise and lower)

Voltage regulators shall be approximately compensated for their internal regulation to provide the specified voltage range at rating in kVA with an 80% lagging power factor load.

5.2.2 Preferred ratings

Preferred ratings of step-voltage regulators shall be based on operation at a frequency of 60 Hz or 50 Hz and nominal system voltages as given in Table 3, Table 4, Table 5, and Table 6.

Table 3—Preferred ratings for liquid-immersed 60 Hz step-voltage regulators (single phase)

Nominal system voltage	BIL (kV)	kVA	Line amperes
2400/4160Y	60	50	200
		75	300
		100	400
		125	500
		167	668
		250	1000
		333	1332
		416	1665
4800/8320Y	75	50	100
		75	150
		100	200
		125	250
		167	334
		250	500
		333	668
		416	833
7620/13 200Y	95	38.1	50
		57.2	75
		76.2	100
		114.3	150
		167	219
		250	328
		333	438
		416	546
		500	656
		667	875
		833	1093
		1000	1312
13 800	95	69	50
		138	100
		207	150
		276	200
		414	300
		552	400
		667	483
		833	604
		1000	725
14 400/24 940Y	150	72	50
		144	100
		288	200
		333	231
		432	300
		576	400
		667	463
		833	578
		1000	694
19 920/34 500Y	150	100	50
		200	100
		333	167
		400	201
		667	334
		833	418
		1000	502
34 500	200	173	50
		345	100
		518	150
		690	200

Table 4—Preferred ratings for liquid-immersed 50 Hz step-voltage regulators (single phase)

Nominal system voltage	BIL (kV)	kVA	Line amperes
6600/11430Y	95	33	50
		66	100
		99	150
		132	200
		198	300
		264	400
		330	500
		396	600
		462	700
		528	800
11000	95	55	50
		110	100
		165	150
		220	200
		330	300
		440	400
		550	500
		660	600
		770	700
		880	800
15000/25980Y	150	75	50
		150	100
		225	150
		300	200
		450	300
		600	400
		750	500
		900	600
22000	150	110	50
		220	100
		330	150
		440	200
		660	300
		880	400
33000	170	165	50
		330	100
		495	150
		660	200

Table 5—Preferred ratings for liquid-immersed 60 Hz step-voltage regulators (three phase)

Nominal system voltage	BIL (kV)	Self-cooled		Self-cooled/forced-cooled	
		kVA	Line amperes	kVA	Line amperes
2400/4160Y	60	500	667	625	833
		750	1000	937	1250
		1000	1334	1250	1667
4800	60	500	577	625	721
		750	866	937	1082
		1000	1155	1250	1443
7620/13 200Y	95	500	219	625	274
		750	328	937	410
		1000	437	1250	546
		1500	656	2000	874
		2000	874	2667	1166
		2500	1093	3333	1458
		3000	1312	4000	1750
7970/13 800Y	95	500	209	625	261
		750	313	937	391
		1000	418	1250	523
		1500	628	2000	837
		2000	837	2667	1116
		2500	1046	3333	1394
		3000	1255	4000	1673
14 400/24 940Y	150	500	125.5	625	156.8
		750	188.3	937	235.4
		1000	251	1250	314
		1500	377	2000	502
		2000	502	2667	669
		2500	628	3333	837
		3000	694	4000	926
19 920/34 500Y	150	500	84	625	105
		750	125.5	937	156.8
		1000	167	1250	209
		1500	251	2000	335
		2000	335	2667	446
		2500	418	3333	557
		3000	502	4000	669

Table 6—Preferred ratings for liquid-immersed 50 Hz step-voltage regulators (three phase)

Nominal system voltage	BIL (kV)	Self-cooled		Self-cooled/forced-cooled	
		kVA	Line amperes	kVA	Line amperes
6600/11430Y	95	500	253	625	316
		750	379	937	474
		1000	505	1250	631
		1500	758	2000	1010
11000	95	500	262	625	328
		750	394	937	492
		1000	525	1250	656
		1500	787	2000	1050
15000/25980Y	150	500	111	625	139
		750	167	937	208
		1000	222	1250	278
		1500	333	2000	444
		2000	444	2667	593
		2500	556	3333	741
22000	150	500	131	625	164
		750	197	937	246
		1000	262	1250	328
		1500	394	2000	525
33000	170	500	87	625	189
		750	131	937	164
		1000	175	1250	219

5.2.3 Supplementary voltage ratings

In addition to their rated voltage, as defined in 5.2.2, voltage regulators shall deliver rated kVA output without exceeding the specified temperature rise per Table 2 at the operating voltages given in Table 7.

Voltage regulators with multitapped voltage transformers and/or utility windings may be operated at voltages other than the rated voltage, as specified per the nameplate, and shall deliver rated line amperes without exceeding the temperature limits of Table 2 and as specified per the nameplate.

Table 7—Supplementary voltage ratings for 60 Hz voltage regulators

Number of phases	Rated voltage	Operating voltage
Single phase	7620	7200
	4330	4160
Three phase	5000	4800
	8660	8320
	13 200	12 470
	13 800	13 200

5.3 Supplementary continuous-current ratings

Single-phase step-voltage regulators rated up to 34.5 kV, inclusive, and rated 668 A and below shall have supplementary continuous-current ratings on intermediate ranges of steps as shown in Table 8. Maximum continuous current shall be 668 A.

Table 8—Supplementary continuous-current ratings for single-phase voltage regulators

Range of voltage regulation (%)	Continuous-current rating (%)
10.00	100
8.75	110
7.50	120
6.25	135
5.00	160

Three-phase step-voltage regulators rated up to 13.8 kV, inclusive, and rated 668 A and below shall have supplementary continuous-current ratings on intermediate ranges of steps as shown in Table 9. Maximum continuous current shall be 668 A.

Table 9—Supplementary continuous-current ratings for three-phase voltage regulators

Range of voltage regulation (%)	Continuous-current rating (%)
10.00	100
8.75	108
7.50	115
6.25	120
5.00	130

5.4 Taps

Load tap changing equipment shall be furnished to provide approximately ±10 % automatic adjustment of the unregulated supply on the source side of the voltage regulator. This is to be done in approximately 0.625 percent steps, with sixteen steps above and sixteen steps below rated voltage.

5.5 Operating voltage limits

Voltage regulators, including their controls, shall be suitable for operation within the following limits of voltage provided that the rated load current is not exceeded:

a) A minimum input voltage of 97.75 V times the ratio of voltage transformer.

b) A maximum input voltage at rated load amperes of 1.05 times the rated input voltage of the voltage regulator or 137.5 V times the ratio of voltage transformer, whichever is less.

c) A maximum input voltage at no load of 1.10 times the rated input voltage of the voltage regulator or 137.5 V times the ratio of voltage transformer, whichever is less.

d) A minimum output voltage of 103.5 V times the ratio of voltage transformer.

e) A maximum output voltage of 1.10 times the rated voltage of the voltage regulator or 137.5 V times the ratio of voltage transformer, whichever is less.

f) The output voltage obtainable with a given input voltage is limited also by the voltage regulator voltage range.

Typical examples of the application of these rules to some common ratings of voltage regulators are given in Table 10.

Table 10—Typical examples of operating voltage limits including all operating voltage tolerances

Nominal system voltage	Regulator voltage rating (V)		Voltage supply ratio[a]	Input voltage (V)			Output voltage (V)	
	Single phase	Three phase		Minimum	Maximum at rated-load amperes	Maximum at no-load	Minimum	Maximum at rated-load amperes or at no-load
2400	2500	—	20	1955	2625	2750	2070	2750
2400/4160Y	2500	—	20	1955	2625	2750	2070	2750
2400/4160Y	—	4330	34.6	3380	4550	4760	3580	4760
4800	5000	5000	40	3910	5250	5500	4140	5500
7200	7620	—	60	5870	8000	8250	6210	8250
7200	—	8660	60	5870	8250	8250	6210	8250
4800/8320Y	5000	—	40	3910	5250	5500	4140	5500
8320	—	8660	69.3	6775	9090	9525	7170	9525
11000	11000	—	91.7	8960	11 550	12 100	9490	12 100
6600/11 430Y	6600	—	55	5380	6930	7260	5690	7260
12 470	13 800	13 800	104	10 170	14 300	14 300	10 760	14 300
7200/12 470Y	7620	—	60	5870	8000	8250	6210	8250
7200/12 470Y	—	13 800	104	10 170	14 300	14 300	10 760	14 300
7620/13 200Y	7620	—	63.5	6210	8000	8380	6570	8380
7620/13 200Y	7620	—	66.3	6480	8000	8380	6860	8380
7620/13 200Y	—	13 200	110	10 750	13 860	14 520	11 400	14 520
7970/13 800Y	—	13 800	115	11 240	14 490	15 180	11 900	15 180
7970/13 800Y	7970	—	66.3	6480	8360	8760	6860	8760
13 200	13 800	13 800	110	10 750	14 490	15 125	11 400	15 125
14 400	13 800	13 800	120	11 730	14 490	15 180	12 420	15 180
22 000	22 000	—	183.3	17 920	23 100	24 200	18 975	24 200
14 400/24 940Y	14 400	—	120	11 730	15 120	15 840	12 420	15 840
15 000/25 980Y	15 000	—	120	11 730	15 750	16 500	12 420	16 500
19 920/34 500Y	19 920	—	166	16 230	20 920	21 910	17 180	21 910
14 400/24 940Y	—	24 940	208	20 330	26 190	27 435	21 530	27 435
33 000	33 000	—	275	26 880	34 650	36 300	28 460	36 300
19 920/34 500Y	—	34 500	287.5	28 100	36 225	37 950	29 760	37 950
34 500	34 500	—	287.5	28 100	36 225	37 950	29 760	37 950

NOTE—Example values are derived using procedures of 5.5.

[a] Where the listed voltage ratio is provided, an ancillary transformer may be required.

5.6 Voltage supply ratios

Values of voltage supply ratios are given in Table 11. When a voltage supply ratio is specified that is not a preferred value shown in Table 11, an ancillary transformer may be furnished in the unit or control to modify the preferred ratio.

Table 11 —Values of voltage supply ratios

Voltage regulator rating (V)		Values of voltage supply ratios
Single phase	Three phase	
2500		20, 20.8
	4330	34.6, 36.1
5000	5000	40, 41.7
6600		55
7620		60, 63.5
7970		66.4
	8660	69.3, 72.2
11 000		91.7
	13 200	110, 104
13 800	13 800	115, 110
14 400		120
15 000		120
19 920		166
22 000		183.3
	24 940	208
33 000		275
34 500	34 500	287.5

5.7 Insulation levels

Voltage regulators shall be designed to provide coordinated applied-voltage and lightning impulse insulation levels on line terminals, and applied-voltage insulation levels on neutral terminals. The identity of a set of coordinated levels shall be its basic impulse insulation level (BIL), as shown in Table 12.

NOTE—When single-phase voltage regulators are connected in wye, the neutral of the voltage regulator bank shall be connected to the neutral of the system. A closed or open delta connection of the voltage regulators is recommended when the system is three-wire ungrounded.

Table 12 —Interrelationships of dielectric insulation levels for voltage regulators used on systems with BIL ratings of 200 kV and below

BIL kV	Applied-voltage insulation level (kV rms)	Impulse levels		
		Full wave	Chopped wave	
		(kV crest)	(kV crest)	Min time to flashover (µs)
60	19	60	66	1.5
75	26	75	83	1.5
95	34	95	105	1.8
110	34	110	120	2.0
150	50	150	165	3.0
170	70	170	187	3.0
200	70	200	220	3.0

5.8 Losses

The losses specified by the manufacturer shall be the *no-load (excitation) losses* and *total losses of a regulator,* as defined in Clause 3.

5.8.1 Total losses

The total losses of a voltage regulator shall be the sum of the no-load (excitation) and load losses.

5.8.2 Tolerance for losses

Unless otherwise specified, the losses represented by a test of a voltage regulator shall be subject to the following tolerances: the no-load losses of a voltage regulator shall not exceed the specified no-load losses by more than 10%, and the total losses of a voltage regulator shall not exceed the specified total losses by more than 6%. Failure to meet the loss tolerances shall not warrant immediate rejection but lead to consultation between purchaser and manufacturer about further investigation of possible causes and the consequences of the higher losses.

NOTE—Since losses will differ at different operating positions of the voltage regulator, care must be exercised in the consideration of tap position with losses. Some styles of step-voltage regulators will exhibit appreciable change in load loss when boosting versus bucking, or will exhibit appreciable change in no-load loss on alternate tap positions. See 5.8.3.

5.8.3 Determination of losses and excitation current

No-load (excitation) losses and exciting current shall be determined for the rated voltage and frequency on a sine-wave basis, unless a different form is inherent in the operation of the apparatus.

Load losses shall be determined for rated voltage, current, and frequency and shall be corrected to a reference temperature equal to the sum of the limiting (rated) winding temperature rise by resistance from Table 2 plus 20 °C.

Since losses may be very different at different operating positions and with various design options, losses shall be considered in practice as the sum of no-load and load losses where:

a) No-load loss is the average of no-load loss in the neutral and next adjacent boost position with rated voltage applied to the shunt or series winding for voltage regulators that do not include a series transformer.

NOTE—It will be apparent, in the case of a Type B step-voltage regulator that is on the next adjacent boost position, that the excitation voltage applied at the source terminal will be higher at the shunt winding. Care must be exercised to assure that rated excitation is present on the shunt winding; this may be accomplished by exciting the voltage regulator from the load terminal.

b) No-load loss is reported for neutral position, maximum boost position, and position adjacent-to-maximum boost position for voltage regulators that include a series transformer.

c) Load loss is the average load loss in both the maximum and adjacent-to-maximum buck positions, and the maximum and adjacent-to-maximum boost positions (that is, four positions) with rated current in the windings.

5.9 Short-circuit requirements

5.9.1 General

Step-voltage regulators shall be designed and constructed to withstand the mechanical and thermal stresses produced by external short circuits of a maximum value of 25 times the base rms symmetrical rated load current to a maximum requirement of 16 kA rms symmetrical.

a) The first-cycle asymmetrical peak current that the voltage regulator is required to withstand shall be determined as shown in Equation (1) and Table 13.

$$ISC(\text{pk asym}) = KISC(\text{rms sym}) \tag{1}$$

Table 13—Values of K

Base rated kVA	K	
	60 Hz	**50 Hz**
< 165 (single phase)	2.26 [a]	2.19 [a]
≥ 165 (single phase) ≥ 500 (three phase)	2.60 [b]	2.55 [b]

[a] Value of the first-cycle asymmetrical peak current is based on an X/R ratio of 6 and 5, 60 and 50 Hz, respectively, which are common for distribution circuits.

[b] Value of the first-cycle asymmetrical peak current is based on an X/R ratio of 17 and 14, 60 and 50 Hz, respectively, which are common for substation circuits.

b) The short-circuit current shall be assumed to be a duration of 2 s to determine the thermal stresses.

It is recognized that short-circuit withstand capability can be adversely affected by the cumulative effects of repeated mechanical and thermal overstressing, as produced by short circuits and loads above the nameplate rating. Since means are not available to continuously monitor and quantitatively evaluate the degrading effects of such duty, short-circuit tests, when required, should be performed prior to placing the voltage regulator in service.

Voltage regulator components such as leads, bushings, and load tap changers that carry current continuously shall comply with all the requirements of 5.9.1. Load tap changers are not required to change tap position coincident with a short-circuit condition.

It is recommended that current-limiting reactors be installed by the user, where necessary, to limit the short-circuit current to a maximum of 25 times the base rated full-load current or 16 000 A, whichever is less.

NOTE 1—Larger kVA sizes for the same voltage rating should be considered if the available fault current exceeds the 25 times base rated current.

NOTE 2—User may specify a larger short-circuit withstand value due to unique system parameters. An example as such is a short-circuit withstand of 40 times the base rated load current or 20 000 A whichever is less. Application, limitations, design, and resulting cost are to be agreed upon by the user and the manufacturer.

5.9.2 Mechanical capability demonstration

It is not the intent of this subclause that every voltage regulator design is short-circuit tested to demonstrate adequate construction. When specified, tests of short-circuit mechanical capability shall be performed as described in 8.8.

5.9.3 Thermal capability of voltage regulators for short-circuit conditions

The temperature of the conductor material in the windings of voltage regulators under the short-circuit conditions specified in 5.9.1, as calculated by methods described in 8.9.4, shall not exceed 250 °C for a copper conductor or 200 °C for an electrical conductor (EC) aluminum. A maximum temperature of 250 °C shall be allowed for aluminum alloys that have resistance to annealing properties at 250 °C, equivalent to EC aluminum at 200 °C, or for application of EC aluminum where the characteristics of the fully annealed material satisfy the mechanical requirements. In setting these temperature limits, the following factors were considered:

a) Gas generation from fluid or solid insulation
b) Conductor annealing
c) Insulation aging

5.10 Tests

Tests are divided into two categories: routine and design. Routine tests are made for quality control by the manufacturer to verify during production that the product meets the design specifications. Design tests are made to determine the adequacy of the design of a particular type, style, or model of equipment or its component parts. Design adequacy includes but is not limited to: meeting assigned ratings, operating satisfactorily under normal service condition or under special conditions if specified, and compliance with appropriate standards of the industry.

5.10.1 Routine tests

Routine tests *shall* be made on all voltage regulators per the following list:

a) Resistance measurements of all windings (see 8.1)
b) Ratio test on all tap connections (see 8.2)
c) Polarity test (see 8.3)
d) Operational test of all devices. Controlled devices such as load tap changers, position indicators, fans, pumps, etc., shall be operated for proper functioning.
e) Leak test
f) No-load (excitation) loss at rated voltage and rated frequency (see 8.4)
g) Excitation current at rated voltage and rated frequency (see 8.4)
h) Impedance and load loss at rated current and rated frequency (see 8.5)
i) Lightning impulse test (see 8.6.3)
j) Applied-voltage test (see 8.6.5)
k) Induced-voltage test (see 8.6.6)
l) Insulation power factor test (see 8.6.7)
m) Insulation resistance test (see 8.6.8)